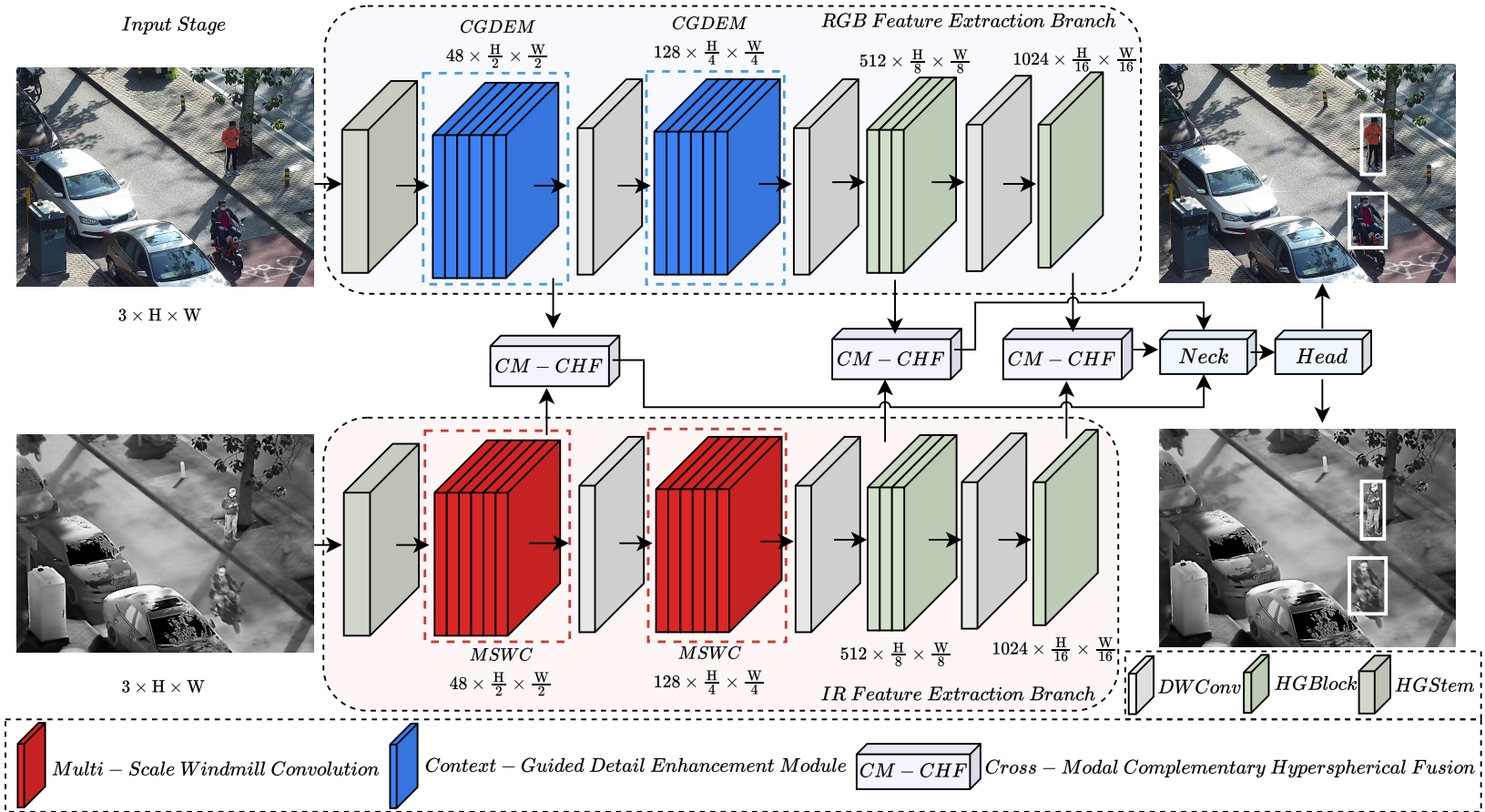
Input Stage
CGDEM
CGDEM
RGB Feature Extraction Branch
48 × H/2 × W/2
128 × H/4 × W/4
512 × H/8 × W/8
1024 × H/16 × W/16
3 × H × W
CM − CHF
CM − CHF
CM − CHF
Neck
Head
MSWC
MSWC
48 × H/2 × W/2
128 × H/4 × W/4
512 × H/8 × W/8
1024 × H/16 × W/16
IR Feature Extraction Branch
3 × H × W
DWConv
HGBlock
HGStem
Multi − Scale Windmill Convolution
Context − Guided Detail Enhancement Module
CM − CHF
Cross − Modal Complementary Hyperspherical Fusion